

# **Python For DA Module End Assignment**

# Social Media Engagement Analytics Using Python

**Objective:** Analyse a 5,000-record social media dataset to understand which content, audiences and behaviours drive engagement.

**Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly (Google Colab).

**Dataset:** social\_media\_engagement\_5000.csv, with 19 columns: user details (age, gender, country), post details (type, category, hashtags, posted date), metrics (likes, comments, shares, watch time, impressions, followers), device, sentiment, verified status and engagement rate.

## Task 1: Data Import and Setup

- Imported the CSV with `pd.read_csv()`.
- Checked data types with `df.dtypes`.
- Converted `posted_at` to datetime (format dd-mm-yyyy) and `is_verified` to Boolean

```
#Data Import & Setup
import pandas as pd

df=pd.read_csv('/content/social_media_engagement_5000.csv')

df.head()
```

## Task 2: Data Cleaning

```
# Data Cleaning
# Missing values
print(df.isnull().sum())
for c in ["age", "likes", "comments", "shares"]:
    df[c] = df[c].fillna(df[c].median())
df["gender"] = df["gender"].fillna(df["gender"].mode()[0])
df["sentiment"] = df["sentiment"].fillna("neutral")
df["posted_at"] = df["posted_at"].ffill().bfill()

# Duplicates (user_id repeats are normal, so check full rows and post_id)
print(df.duplicated().sum(), df["post_id"].duplicated().sum())
df = df.drop_duplicates().drop_duplicates(subset="post_id").reset_index(drop=True)

# Standardize categories
for c in ["gender", "post_type", "post_category", "country", "device_type", "sentiment"]:
    df[c] = df[c].astype(str).str.strip().str.title()

# Unrealistic values
df = df[(df["likes"] >= 0) & (df["comments"] >= 0) & (df["shares"] >= 0)]
df = df[(df["age"] >= 13) & (df["age"] <= 100)]
df["total_interactions"] = df["likes"] + df["comments"] + df["shares"]
print("Rows with interactions > impressions:", (df["total_interactions"] > df["impression_count"]).sum())
df = df[df["total_interactions"] <= df["impression_count"]] # engagement_rate must be <= 1
df["engagement_rate"] = df["total_interactions"] / df["impression_count"] # recompute after imputing

# Feature cleaning
df["hashtag_count"] = df["hashtags"].fillna("").str.count("#")
df["sentiment"] = df["sentiment"].replace({"": "Neutral", "Nan": "Neutral"})
df = df.reset_index(drop=True)
```

| Step               | Action                                                                                                                                                                               |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Missing values     | 150 each in age, gender, likes, comments, shares, sentiment. Numeric columns filled with the median, gender with the mode, sentiment with "Neutral"                                  |
| Duplicates         | 0 duplicate rows; 9 duplicate <code>post_ids</code> removed                                                                                                                          |
| Standardisation    | Stripped spaces and applied consistent title case to gender, post type, category, country, device and sentiment                                                                      |
| Unrealistic values | Removed negative values and ages outside 13-100. Removed 599 rows where likes + comments + shares exceeded impressions (engagement rate above 100%), then recomputed engagement rate |
| Feature cleaning   | Created <code>hashtag_count</code> by counting "#" in hashtags                                                                                                                       |

### Task 3: Data Exploration

- Used `head()`, `tail()`, `shape`, `columns`, `info()`, `dtypes` and `describe()`.
- Used `value_counts()`, `unique()` and `nunique()` on categorical columns. Sentiment: Positive 2,085, Neutral 1,479, Negative 828.
- Correlation matrix: engagement rate vs impressions is **-0.69** (strongest), and engagement rate vs likes is **0.44**. Likes vs impressions is 0.12. All other pairs are near 0.
- `groupby()` summaries: average likes by post type and total impressions by country.

```
#Data Exploration
print(df.head(), df.tail(), df.shape, df.columns, sep="\n")
df.info(); print(df.dtypes)
print(df.describe())
for c in ["gender", "country", "post_type", "post_category", "device_type", "sentiment"]:
    print(df[c].value_counts(), df[c].unique(), df[c].nunique(), sep="\n")
num = ["age", "likes", "comments", "shares", "watch_time_sec", "impression_count",
       "follower_count", "engagement_rate", "hashtag_count"]
corr = df[num].corr()
print(corr)
print(df.groupby("post_type")["likes"].mean())
print(df.groupby("country")["impression_count"].sum())
```

## Task 4: Data Wrangling

- Created `engagement_score = likes + 2×comments + 3×shares, log_impressions, log_likes, day_of_week` and `month`.
- Used `merge()` to join country-level average engagement rate back to the main table.
- Ran groupby summaries by post type, country and sentiment.

```
#Data Wrangling
df["engagement_score"] = df["likes"] + 2*df["comments"] + 3*df["shares"]
df["log_impressions"] = np.log1p(df["impression_count"])
df["log_likes"] = np.log1p(df["likes"])
df["day_of_week"] = df["posted_at"].dt.day_name()
df["month"] = df["posted_at"].dt.to_period("M").astype(str)

# merge demo: country average engagement joined back
country_avg = df.groupby("country", as_index=False)["engagement_rate"].mean() \
    .rename(columns={"engagement_rate": "country_avg_er"})
df = df.merge(country_avg, on="country", how="left")

cols = ["likes", "comments", "shares", "impression_count", "engagement_rate", "engagement_score"]
for g in ["post_type", "country", "sentiment"]:
    print(df.groupby(g)[cols].mean().round(3), "\n")
```

## Task 5: Statistical Analysis

```
#Statistical analysis
cols = ["likes", "comments", "shares", "watch_time_sec", "engagement_rate", "follower_count"]
stats = df[cols].agg(["mean", "median", "std", "var", "skew", "kurt"]).T
stats["mode"] = [df[c].mode()[0] for c in cols]
stats = stats.join(df[cols].quantile([.25, .5, .75, .9]).T.rename(
    columns={.25: "p25", .5: "p50", .75: "p75", .9: "p90"}))
print(stats)
```

| Column   | Mean  | Median | Std   | Skew | Kurtosis |
|----------|-------|--------|-------|------|----------|
| likes    | 9,748 | 9,880  | 5,681 | 0.06 | -1.14    |
| comments | 1,491 | 1,497  | 857   | 0.01 | -1.15    |

| Column          | Mean    | Median  | Std     | Skew  | Kurtosis |
|-----------------|---------|---------|---------|-------|----------|
| shares          | 996     | 1,012   | 570     | -0.01 | -1.15    |
| watch_time_sec  | 3,992   | 3,996   | 2,312   | -0.00 | -1.20    |
| engagement_rate | 0.286   | 0.226   | 0.209   | 1.31  | 1.27     |
| follower_count  | 394,537 | 390,900 | 230,714 | 0.03  | -1.19    |

Mode, variance and the 25th, 50th, 75th and 90th percentiles were also computed.

## Task 6: Data Visualization (14 plots)

- **Matplotlib** : scatter (likes vs impressions), line (daily engagement trend), bar (posts by category), pie (gender distribution), histogram (age), box plot (engagement rate).

```
#Matplotlib
plt.scatter(df["impression_count"], df["likes"], alpha=.3)
plt.xlabel("Impressions"); plt.ylabel("Likes"); plt.title("Likes vs Impressions"); plt.show()

df.groupby("posted_at")["engagement_score"].mean().plot(figsize=(10,4), title="Daily Engagement Trend"); plt.show()
df["post_category"].value_counts().plot(kind="bar", title="Posts by Category"); plt.show()
df["gender"].value_counts().plot(kind="pie", autopct="%1.1f%%", title="Gender Distribution"); plt.ylabel(""); plt.show()
plt.hist(df["age"], bins=20); plt.title("Age Distribution"); plt.show()
plt.boxplot(df["engagement_rate"]); plt.title("Engagement Rate"); plt.show()
```

- **Seaborn** : count plot (post type), bar plot (avg likes by category), violin (followers vs sentiment), pair plot, heatmap (correlation), swarm plot (engagement vs device).

```
#Seaborn
sns.countplot(data=df, x="post_type"); plt.show()
sns.barplot(data=df, x="post_category", y="likes"); plt.xticks(rotation=45); plt.show()
sns.violinplot(data=df, x="sentiment", y="follower_count"); plt.show()
sns.pairplot(df[["likes", "comments", "shares", "impression_count", "engagement_rate"]].sample(500, random_state=1)); plt.show()
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm"); plt.show()
sns.swarmplot(data=df.sample(500, random_state=1), x="device_type", y="engagement_rate"); plt.show()
```

- **Plotly** : interactive bubble chart (likes vs impressions, sized by shares), line chart (daily engagement), bar chart (engagement rate by country)

```
#Plotly|
px.scatter(df, x="impression_count", y="likes", size="shares", color="post_type",
           title="Likes vs Impressions (bubble = shares)").show()
daily = df.groupby("posted_at", as_index=False)["engagement_score"].mean()
px.line(daily, x="posted_at", y="engagement_score", title="Daily Engagement").show()
px.bar(df.groupby("country", as_index=False)["engagement_rate"].mean(),
       x="country", y="engagement_rate", title="Avg Engagement Rate by Country").show()
```

## Final Insights

### Content Performance

- **Highest-engagement post type:** Image (0.290), then Text (0.286), Video (0.285) and Reel (0.281). The differences are small.
- **Best category:** Education (0.2945), almost tied with Music (0.2944). Fitness is lowest (0.2777).
- **Top countries by engagement rate:** Germany (0.309), Canada (0.298) and Japan (0.288). France is lowest (0.267).

### User Trends

- **Age:** Engagement falls as age rises: 13-17 (0.309), 18-25 (0.291), 26-35 (0.288), 36-50 (0.284), 51+ (0.276).
- **Verified accounts:** They perform slightly worse than unverified ones. Engagement rate is 0.260 vs 0.288, and average likes are 9,310 vs 9,794.

### Behavioral Insights

- **Best time for impressions:** The data has no hour of day, so day of week was used. Saturday (56,802) and Tuesday (56,800) are highest, and Thursday is lowest (53,990).
- **Device vs watch time:** Mobile (4,080 sec) is highest, then Tablet (3,965) and Desktop (3,930). The differences are small.

### Sentiment Analysis

- **Best sentiment:** Positive (0.288), just ahead of Neutral (0.287).
- **Negative and neutral posts:** Negative posts have the lowest engagement rate (0.277) but slightly more comments (1,513 vs 1,466 for positive), which suggests they provoke discussion. Neutral posts perform almost as well as positive ones.

## Conclusion

Engagement is fairly uniform across post types, categories and sentiments. The clearest patterns are:

1. Younger users engage more.
2. Germany and Canada lead on engagement.
3. Education and Music content slightly outperform other categories.
4. Verified status gives no advantage.
5. Engagement rate is driven mainly by impressions (correlation -0.69)